CALMCARE

The multilingual, voice-first, safety-verified recovery companion

Hackathon solution proposal
Problem statement H2: Discharge Summary to Patient-Understandable Care Plan

A complete website concept, patient journey, safety design, evaluation plan and explanation of all 33 proposed features.

Our central promise

Help patients and families understand what their discharge instructions mean, what they need to do next, and when they need help—while preserving the exact medical instructions provided in the source document.

Important scope note: this document explains the official problem statement supplied for H2 and our proposed solution. The 33 features are our design proposal, not a claim that every feature is required by the organisers or that no similar feature exists elsewhere.

Document guide

Use this document as the complete reference for the website idea, team planning and hackathon presentation.

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Quick summary

CalmCare accepts a discharge summary as a PDF, image or camera capture; extracts and checks its contents; turns the verified information into plain-language care instructions; offers multiple supported Indian languages with calm voice playback; and provides carefully controlled schedules, reminders, follow-up support and printable materials. It must never invent a medical instruction to fill a gap.

1. Problem statement explained

Official problem title: H2. Discharge Summary to Patient-Understandable Care Plan

A patient may leave hospital carrying a discharge summary filled with medical abbreviations, unfamiliar words, medicine dosages and follow-up instructions. The patient may not understand what the illness means, how to follow the prescribed plan or which warning signs need urgent attention. The challenge is greater when the patient cannot comfortably read English or has difficulty reading medical documents.

When instructions are misunderstood, forgotten or incorrectly followed, the patient may miss medicines or appointments and may need avoidable medical attention. The proposed system should help patients and families understand the instructions without changing what the clinician prescribed.

The core problem in one sentence

Convert a clinical discharge summary into a faithful, plain-language care plan that a patient and family can understand and use—without adding, dropping or altering any medical instruction.

Why this is not just a summarisation task

- **Completeness:** every important medicine, dose, frequency, duration and follow-up date must be preserved.
- **Correctness:** the simplified version must not change a value or introduce unsupported medical advice.
- **Understanding:** the result should be usable by a person with little medical knowledge and roughly Class 8 reading ability.
- **Accessibility:** the patient may prefer a regional language, spoken guidance, larger text or help from a caregiver.
- **Practicality:** the output should become a real daily-use plan, including a printable one-page fridge sheet.

The supplied problem statement assumes the source discharge summary is clinically correct and complete, and says the system must not infer beyond it. It also states that full electronic health record access is not available. Our design follows those boundaries.

2. What the hackathon problem statement asks us to deliver

The source problem statement supplies data and an evaluation harness. The deliverables below are the explicit requirements from that document; additional CalmCare ideas are enhancements proposed by our team.

PS requirement	Meaning for our solution	How CalmCare will address it
Structured care plan	Organise medicines, warning signs, diet/activity and follow-up appointments.	Separate, labelled sections with clear actions, dates and source references.
Faithfulness over fluency	No dropped medicine, altered dose or invented instruction; hard penalty for these errors.	Structured extraction, rule checks, source traceability, discrepancy flags and human review where needed.
English + one Indian language	The base PS asks for English plus one chosen Indian language, with faithfulness checked in translation.	Minimum compliance: English plus one selected language. Our proposed extension allows multiple selectable supported Indian languages.
Readability score	Report a readability measure for the generated patient-facing output.	Show score and reading-level estimate, with a target around Class 8 where the metric supports the language.
One-page fridge sheet	Produce a practical page the family can keep and refer to.	Printable PDF with only verified key instructions, medicine schedule and follow-up details.
Stretch: spoken audio	Provide an audio version for patients who cannot read.	CalmCare voice playback, selected-language speech and section replay.
Stretch: 7-day timetable	Create a medicine timetable when the source contains sufficient timing information.	Generate only from explicit, verified prescription details; never guess missing times.

Provided resources in the problem statement

- 2,000 synthetic discharge summaries covering diabetes, post-operative recovery, cardiac, respiratory and maternity cases.
- A medical abbreviation glossary and a drug reference table with generic name, common brand and dose form.
- 50 human-written plain-language reference versions.
- An evaluation harness that checks whether source medicines, doses, frequencies and follow-up dates appear in the output.

We will use these resources for testing and evaluation. The drug table can help interpret names and forms, but it must not be used to invent missing prescriptions or override the source summary.

3. Our proposed solution: CalmCare

CalmCare is a patient-focused website that turns a hospital discharge document into a verified, understandable and accessible recovery companion. It combines document extraction, plain-language explanation, multiple language choices, calm speech, instruction verification and optional follow-up support.

What makes our proposed approach distinctive

- **Traceable:** important instructions can be linked back to the exact source passage.
- **Patient-paced:** a calm voice explains one topic at a time and can repeat or simplify it.
- **Safety-gated:** unclear doses, conflicting instructions and uncertain OCR results are flagged rather than guessed.
- **Multilingual:** the patient can choose from the languages actually supported and validated by the speech/translation components.
- **Practical:** verified instructions can become a medicine timetable, reminders, appointment list and one-page fridge sheet.
- **Continuity-aware:** updated discharge summaries can be compared with an earlier plan so changes are not missed.

CalmCare is not a diagnostic tool, a replacement for a doctor or a source of new treatment instructions. It explains and organises what is documented. When the document is unclear, the safe outcome is to ask for clarification.

Primary users

User	Need
Patient	Understand their condition and instructions in everyday words, preferred language and preferred format.
Family caregiver	Help the patient follow the plan, with the patient's permission and clear privacy controls.
Clinician or reviewer (optional workflow)	Resolve unclear or conflicting source information before a plan is marked verified.
Hackathon evaluator	See source-to-output traceability, safety checks, readability results and test outcomes.

4. Website design and complete patient journey

Page 1 — Welcome and accessibility preferences

The first screen explains CalmCare in simple language. The user chooses the language they want to hear and read, can choose more than one output language, sets text size and voice speed, and selects patient or caregiver mode. Language selection is available by typing or speaking when speech recognition supports the language.

Page 2 — Add the discharge summary

Provide two equally visible options in the same upload area: Upload a file and Take a photo / Use camera. Accept PDF, JPG and PNG. Support multiple pages or camera captures, preview, rotate, crop, retake, reorder and remove pages. Ask the user to confirm that all pages have been captured.

Page 3 — Image quality and text extraction

Run OCR for images and scanned PDFs; extract text directly from digital PDFs where possible. Detect pages that appear blurry, cut off, shadowed or unreadable. Highlight uncertain words and values, especially medicine names, doses, units, frequency and dates. Ask for a clearer image or manual confirmation rather than silently guessing.

Page 4 — Review extracted medical facts

Show a structured review of patient-independent facts found in the document: condition or reason for admission as written, medicines, dose, unit, route if stated, frequency, duration, warnings, diet/activity restrictions, follow-up date and contact/location if stated. Each fact should link to its source text. Mark missing, uncertain or conflicting values.

Page 5 — Build the understandable care plan

After extraction and checks, create sections for ‘What your summary says’, ‘Your medicines’, ‘What to watch for’, ‘Food and activity instructions’, and ‘Follow-up’. Explain medical terms using familiar words. Use a calm and hopeful tone without promising recovery or adding unsupported instructions.

Page 6 — Language, text and voice

The user can choose English and one or more supported Indian languages. Display text and play spoken guidance in the selected language. Provide pause, replay, slower speech, section-by-section playback, simpler explanation and a question box. Show the speech-recognition transcript before using a voice command for important actions.

Page 7 — Medicines and daily timetable

Show a visual timetable only where the prescription provides enough information to place a dose in time. If the summary says only ‘twice daily’ without specifying exact clock times, do not invent times. Ask for clarification or show the prescribed frequency without assigning an unapproved schedule.

Page 8 — Reminders and appointments

Let the user opt into reminders based on verified medicine timing, documented follow-up dates and explicit care tasks. Offer in-app/browser/mobile notifications and optional external messaging only where available and consented to. Provide quiet hours, snooze, completion and reschedule controls.

Page 9 — Ask CalmCare and check understanding

The user asks questions by typing or speaking. Answers must be grounded in the verified plan or approved reference content. If the answer is not in the source, say so and recommend checking with the care team. Optional teach-back questions help reveal which instruction needs a clearer explanation.

Page 10 — Printable plan and progress

Offer a one-page fridge sheet, a full care-plan view, a readability report and a safety/faithfulness report. The progress dashboard shows only user-confirmed actions and patient-reported check-ins; it must not claim to prove clinical recovery.

5. All 33 proposed features explained

The feature list below consolidates our discussions. Some features are core requirements, some are safety controls, and some are optional extensions. The numbering preserves our agreed list of 1–28 plus the five latest additions.

1. Discharge-summary upload and camera capture

Users can upload PDF/JPG/PNG files or capture document pages directly through the website camera. Support multi-page capture, preview, rotate, crop, retake and page ordering. Flag blur, glare, cut-off edges and missing pages before extraction.

2. Smart medical information extraction

OCR and document parsing identify medicine names, doses, units, frequencies, durations, warning signs, restrictions and follow-up details. A medical abbreviation glossary can help interpret terms. Extraction confidence is retained so uncertain facts are visible.

3. Plain-language illness explanation

Explain what the discharge summary says about the patient's condition using everyday language for someone with no medical background. Explain medical words simply, avoid unnecessary jargon and do not infer diagnoses or causes not stated in the source.

4. Structured care plan

Organise the source into medicines, warning signs, diet, activity, wound-care or other documented instructions, and follow-up appointments. Each item is traceable to source text; missing or ambiguous information remains flagged.

5. Multiple selectable languages

Provide English plus a choice of supported Indian languages, with one or more selected outputs. The exact available list depends on the validated translation and speech engines. Users can change language without uploading the document again.

6. Voice and text interaction

Patients can type or speak to select language, ask questions or request a section. Show the recognized transcript and allow editing before consequential actions. Keep text input available when microphone access or speech recognition is unavailable.

7. CalmCare spoken guidance

Read the care plan in a warm, patient, calm voice. Speak slowly, use pauses, explain one topic at a time, pronounce dates and numbers clearly, and offer section playback, replay, pause and speed controls.

8. Medical-instruction verification engine

Compare extracted and generated facts for every medicine, dose, unit, frequency, duration and follow-up date. Flag omissions, altered values and unsupported additions. A plan must not be marked verified while a critical discrepancy remains unresolved.

9. Low-literacy and accessibility mode

Use short sentences, large readable type, high contrast, clear icons, keyboard navigation and screen-reader-friendly labels. Keep each screen uncluttered and offer text and audio together.

10. Visual medicine timetable

Present verified medicine instructions in an easy-to-scan schedule. Create exact clock-time slots only when supported by explicit instructions or confirmed by an authorised reviewer; never guess timing or change a prescription.

11. Patient and caregiver mode

Allow a patient to involve a trusted caregiver, with clear consent and permission controls. Display the same verified facts to both parties, while respecting the patient's sharing preferences and privacy.

12. One-page fridge sheet and quality report

Generate a print-ready summary with key medicines, documented warning signs and follow-up details. Provide a separate quality report for completeness, source mismatches, unsupported additions, readability and language validation.

13. Ask My Discharge Summary assistant

Provide grounded text/voice Q&A; about the uploaded plan. Answers must come from the source or approved reference material. If information is absent or unclear, the assistant must say it cannot determine the answer from the document.

14. Teach-back understanding check

Offer gentle questions such as asking the patient to identify their documented follow-up date. If they are unsure, re-explain the relevant instruction in simpler language or audio. It is an understanding aid, not a diagnostic or grading tool.

15. Uncertainty and contradiction detection

Detect low-confidence OCR, incomplete instructions and conflicting medication entries or dates. Show the relevant source passages and request a clearer image, patient confirmation or clinical review instead of resolving uncertainty by guesswork.

16. Empathetic communication mode

Use respectful, non-judgmental language. Avoid blame, frightening phrasing and guaranteed recovery claims. Recognise worry without pretending to know the patient's feelings or giving unsupported reassurance.

17. Follow-up and appointment organiser

Extract documented appointment dates, times and locations. Let users confirm and optionally export them to a calendar. Never invent a date, location or clinician; ask the user to verify incomplete details.

18. Reminder reliability and missed-task handling

Let users configure quiet hours, snooze and reminder frequency. Lack of response is not proof that a dose was missed. Do not advise doubling a missed dose; direct medicine-specific questions to the clinician or pharmacist.

19. Clinical fidelity dashboard

Report whether critical source details were preserved, whether any unsupported instructions appeared, what remains uncertain, and how each language version performed. Avoid a single misleading score that hides a critical error.

20. Low-bandwidth and print-friendly access

Keep core screens lightweight, offer downloadable/printable output and, where technically feasible, allow a previously saved plan to be viewed offline with suitable device privacy controls.

21. Smart notifications and reminders

Offer optional reminders for explicit prescribed medicine timings, documented appointments and documented care tasks. Include taken/snooze/skipped actions, consent-based channels and voice reminders where supported. Only user-confirmed actions count as completed; reminders must not generate medical advice.

22. Daily recovery check-in

Allow users to report how they feel, whether they followed documented tasks and any concerns. Show trends as self-reported information, not a diagnosis. Potentially urgent symptoms should trigger appropriate urgent-care guidance without delaying for a long questionnaire.

23. Personal recovery progress dashboard

Summarise user-confirmed medicine actions, appointments, completed documented tasks and patient-reported check-ins. Clearly state that these records do not prove clinical recovery or replace a medical assessment.

24. Warning-sign guidance

Present warning signs explicitly documented in the discharge summary in clear language. If a user reports potentially urgent symptoms, direct them to appropriate urgent medical help; do not let a chatbot conversation delay care.

25. Condition-specific recovery guidance

Support the provided case categories—diabetes, post-operative, cardiac, respiratory and maternity—by organising relevant source instructions. Any supplementary clinical content must be reviewed and approved; do not create condition-specific advice from model guesswork.

26. Explain this again

Place a clear control beside each instruction to repeat it, slow it down, translate it or explain it more simply. Preserve the original clinical meaning and keep the exact prescribed values visible.

27. Discharge-instruction change management

Compare a newly uploaded summary with the previous verified plan. Highlight additions, removals and changed doses/dates; identify affected reminders and mark uncertain changes for confirmation before replacing the old plan.

28. Privacy-first processing

Collect only necessary information, protect transfers and stored plans, avoid sensitive information in application logs, provide clear deletion controls and use explicit consent for caregiver sharing or optional notifications. The prototype should not require full EHR access.

29. Source-proof for every medical instruction

Let the patient or reviewer tap a critical instruction to see the exact source sentence supporting it. If no reliable source passage exists, do not present the instruction as source-verified.

30. Adaptive understanding engine

Use optional teach-back answers, requests to repeat and patient feedback to select a clearer explanation style: simpler wording, slower speech, a visual, or another phrasing. Adapt the explanation, never the prescription.

31. Discharge change impact analyser

When a new document is uploaded, identify which medicines, dates, tasks and reminders may be affected. Flag obsolete reminders for review and do not silently apply ambiguous changes.

32. Cross-language meaning consistency checker

Compare each translation with the verified source facts, particularly medicine names, numbers, units, frequencies, dates, warnings and restrictions. Flag a language version if a critical detail changes or disappears.

33. Patient-specific instruction conflict detector

Find contradictions inside one discharge summary, such as two different doses or follow-up dates. Show both source passages and ask for confirmation or clinical review rather than automatically choosing one.

6. CalmCare AI: voice and conversation design

The voice experience is central to our idea. The AI should not sound like a machine reading a hospital document. It should act like a patient, respectful guide who explains one point at a time and gives the patient control over the pace.

Voice behaviour requirements

- Start at a slow, clear pace with natural pauses between instructions.
- Use short, familiar sentences and explain medical terms before using them repeatedly.
- Pause after a medicine, date or important warning; offer the patient a chance to ask a question.
- Support “Speak slower”, “Repeat”, “Explain more simply”, “Pause”, “Replay this section” and “Let me ask a question”.
- Allow voice interruption where the platform supports it, and retain a visible text version of the explanation.
- Use a language-specific voice with good pronunciation for numbers, dates and medicine names; validate the actual language and voice provider before promising support.
- Avoid scolding, pressure, exaggerated cheerfulness, fear-based language and promises that everything will be fine.
- If speech recognition is uncertain, show the transcript and ask the user to confirm it before changing settings or recording a response.

Example voice script

> *“Hello. We can go through your care instructions together, one step at a time. You do not need to remember everything at once. I will explain what your discharge summary says, and we can repeat any part you want. Would you like to start with your medicines or your follow-up appointment?”*

The script is an example of tone only. Patient-specific explanations must use the actual discharge summary and must not add a diagnosis, medicine instruction or clinical promise.

Voice input and voice output are different capabilities

Voice input means speech-to-text: the patient speaks and the system transcribes the request. Voice output means text-to-speech: CalmCare reads the verified explanation aloud. The website should support both where the selected language is technically available, and always provide a text alternative.

7. Safety and medical-fidelity rules

The main risk is not whether the output sounds fluent. The main risk is whether it changes or invents medical instructions. Safety checks therefore need to be part of the workflow, not an optional decoration.

Rule	Required behaviour
No inference beyond source	The source is assumed clinically correct and complete, but CalmCare must not fill gaps with its own guesses. If an instruction is absent, label it absent or ask for confirmation.
Hard-stop critical mismatch	An altered dose, missing medicine, unsupported instruction or critical translation mismatch prevents the affected plan from being labelled verified.
Source traceability	Store the exact source passage for each critical fact wherever technically possible. Make uncertainty visible to the patient/reviewer.
OCR uncertainty	Blurry or ambiguous numbers, decimal points, units, medicine names and dates must be re-captured or reviewed.
Conflicting entries	Display both conflicting passages. Do not choose the entry that looks more likely.
Reminder generation	Use only a verified schedule. If the source does not provide clock times, do not invent clock times.
Missed dose	Record only what the patient confirms. Never advise a catch-up or double dose; refer to the written instructions or clinician/pharmacist.
Urgent symptoms	For potentially urgent symptoms, direct the patient to appropriate urgent medical help without delaying through routine Q&A.;
Translation safety	Validate critical numbers, units, frequencies, dates, restrictions and warnings in each language. Do not assume fluent translation is faithful translation.
Privacy and consent	Use minimal data, secure processing, clear retention/deletion controls and explicit consent for sharing or optional messaging.

Verification states for the website

State	Meaning
Extracted — needs review	The document has been read, but one or more facts may be uncertain or conflicting.
Ready for validation	A structured draft exists and is undergoing source comparison and language checks.
Verified against source	Critical fields passed the configured checks and no unresolved critical mismatch remains. This does not mean a clinician independently certified the medical correctness of the source.
Review required	A conflict, low-confidence critical field or translation discrepancy remains. Do not activate affected reminders until resolved.

8. System architecture and technology plan

Recommended prototype architecture

Layer	Responsibility	Possible implementation
Website interface	Upload/camera, review, language controls, voice controls, plan, reminders and print view.	Streamlit for fastest demo; or React for a more polished responsive website.
Document ingestion	PDF text extraction, image preprocessing, page ordering and quality warnings.	PyMuPDF for digital PDFs; Pillow/OpenCV for image preprocessing; Tesseract or a suitable OCR service.
Medical extraction	Convert document text into a structured record with source spans and confidence/uncertainty markers.	Python + validated JSON/Pydantic schema; LLM extraction constrained by schema plus deterministic checks.
Simplification engine	Explain source facts in plain language without adding instructions.	LLM with strict source-grounded prompts, structured outputs and glossary support.
Verification layer	Compare source facts and output; validate medication fields, dates, additions, omissions and translations.	Python rule-based checks, exact field comparison, date/number/unit checks and evaluator integration.
Language and voice	Translation, speech input, speech output and playback controls.	Choose providers based on verified support for target languages, pronunciation, latency, cost and privacy.
Outputs	Fridge sheet, readability and safety report, timetable and optional calendar export.	ReportLab PDF, readability tooling appropriate to language, calendar file/export where needed.
Data and privacy	Session state, consent, minimal storage, deletion and optional reminders.	Start with temporary session storage; use protected persistence only when needed.

Recommended processing pipeline

Step	Process
1. Capture	Upload a file or capture every page with the camera.
2. Read	Extract PDF text or run OCR; assess image quality.
3. Structure	Extract medicines, values, dates, warning signs and restrictions with source locations.
4. Review uncertainty	Ask for recapture or confirmation when critical text is unclear or contradictory.
5. Simplify	Generate patient-friendly explanations from verified source facts.
6. Validate	Compare the care plan and each translation against the structured source record.
7. Deliver	Show text, play calm voice, create a valid timetable, reminders and fridge sheet.
8. Support	Offer grounded Q&A;, teach-back, caregiver sharing and change detection.

9. Evaluation plan and success measures

Use the provided 2,000 synthetic summaries, glossary, drug table, 50 human-written reference versions and evaluation harness. Do not invent results before testing. The prototype should report actual measured performance.

Metric	What to test	How to report
Medicine preservation	Whether every source medicine is present in output.	Counts and rates of omissions; separate critical errors.
Dose/unit preservation	Whether dose values and units match exactly.	Mismatch count by type: dose, unit, decimal, route where stated.
Frequency and duration	Whether prescribed frequency/duration is retained without changes.	Exact-match or structured-field accuracy; list failure cases.
Follow-up fidelity	Whether dates and documented appointment details are preserved.	Date extraction and output match; missing/altered dates.
Unsupported additions	Whether output includes instructions absent from source or approved content.	Count of unsupported additions; classify severity.
Translation consistency	Whether critical fields are preserved in each selected language.	Per-language error counts, not just an average.
Readability	Whether patient-facing language approaches the target reading level.	Report metric, tool, language and limitations.
OCR reliability	How often critical fields are extracted correctly from PDF/image/camera input.	Separate digital PDFs, scans, blurry photos and multi-page captures.
Patient comprehension	Whether users can answer basic questions about the plan after explanation.	Small usability study or synthetic task-based checks; do not claim clinical effectiveness from a demo.
Voice usability	Whether selected-language playback is understandable and controllable.	User feedback on pace, pronunciation, replay and speech recognition.
Change detection	Whether updated summaries reveal altered doses, dates and instructions.	Test cases with known changes and conflicts.

Suggested demo test cases

- A clear digital PDF with complete medication details.
- A scanned page with a blurry dose that should be flagged rather than guessed.
- Two conflicting medicine doses in one document.
- A follow-up date that must appear unchanged in the output.
- A translation test where a number or frequency is intentionally corrupted to ensure the checker catches it.
- A new discharge summary that changes one medicine and should trigger reminder review.
- A patient asking a question that is not answered by the document; CalmCare should admit the information is unavailable.

10. Build roadmap, demo flow and pitch

Build in layers. A reliable small prototype is stronger than a large demo with unsafe medicine generation.

Phase	Build	Definition of done
Phase 1 — Core	Upload and camera capture; OCR; structured extraction; plain-language care plan; English + one supported Indian language; source comparison; fridge sheet.	Demonstrate one full case and pass core medicine/date checks.
Phase 2 — Safety	Critical-field confidence, source proof, conflict detection, translation checks, review states and clear error reports.	Show that blurry, missing and conflicting details are flagged instead of guessed.
Phase 3 — CalmCare voice	Speech input, transcript review, calm speech output, pause/replay/slower controls, section-level explanation.	Demonstrate supported language with understandable pronunciation and text fallback.
Phase 4 — Practical support	Validated medicine timetable, appointment organiser, reminders, caregiver mode, teach-back and change comparison.	Show that reminders use verified information and updated instructions affect old reminders safely.
Phase 5 — Polish	Accessibility, privacy/deletion controls, low-bandwidth/print view and evaluation dashboard.	Run test cases and present measured results and known limitations.

Recommended live demo (5–7 minutes)

- Introduce the patient problem using one example of a dense discharge instruction.
- Upload a PDF and capture an extra page through the camera option.
- Show OCR uncertainty or source review so the audience sees the system does not blindly trust extraction.
- Generate a plain-language plan and switch to a selected Indian language.
- Play CalmCare's slow, patient voice; ask it to repeat or explain one instruction more simply.
- Tap an instruction to reveal its source sentence and show the fidelity check.
- Demonstrate a conflicting dose or changed summary and show that the system flags it.
- Generate the one-page fridge sheet and show a reminder based only on verified timing.

Short pitch

“CalmCare turns hospital discharge summaries into understandable, multilingual recovery plans. Patients can upload or photograph their document, hear a calm explanation in a language they choose, and see medicines and follow-up instructions in a clear format. Unlike a simple summariser, CalmCare checks every critical instruction against the source, links instructions back to the original text and flags uncertainty instead of guessing. The goal is to make discharge instructions easier to understand while protecting their medical meaning.”

11. Example patient journey

This is a fictional interaction showing the intended user experience. It is not a medical recommendation.

Stage	What the patient sees or hears	What the system must do
Upload	“Upload your discharge summary” or “Take a photo”.	Accept supported file/camera input, show page preview and warn about unreadable pages.
Review	“We could not read this dose clearly. Please retake this area or ask your care team to confirm it.”	Keep the dose unverified; do not infer the likely number.
Explanation	“Your summary says you have diabetes. I can explain the instructions it contains, one step at a time.”	Explain only source-supported facts; no unsupported diagnosis or promise.
Medicine	“This is the medicine and dose written in your document. Tap here to see the original sentence.”	Preserve name, dose, unit, frequency and duration; link to source.
Voice	“Would you like me to repeat that more slowly?”	Pause/replay controls; selected language only if supported.
Question	Patient asks a question not answered in the summary.	Say the document does not provide that information and suggest checking with the clinician/pharmacist.
Reminder	“Reminder: your documented medicine instruction is due.”	Use only a validated schedule; record completion only after patient confirmation.
New document	“This summary differs from your previous plan. One medicine instruction changed.”	Highlight changes and pause affected reminders pending confirmation/review.
Print	A one-page fridge sheet with verified essentials.	Keep it readable, accurate and traceable; include only supported instructions.

12. Assumptions, limitations and responsible use

Assumptions taken from the supplied problem statement

- The source discharge summary is clinically correct and complete.
- The system must not infer beyond the source summary.
- The target reading level is roughly Class 8.
- The system has no access to the patient's full EHR beyond the discharge summary.
- Structure and simplicity matter more than elegant prose for a patient with low literacy.

Implementation limitations to state honestly

- Camera capture, OCR, speech recognition, text-to-speech and translation quality vary by device, document quality, language and provider. Validate supported languages rather than promising universal support.
- A readability score is a useful signal, not proof that a patient understands the explanation, and common formulas may not work equally well for all Indian languages.
- A rule-based check can catch many critical mismatches but cannot prove clinical correctness in every case. Uncertainty and human review remain important.
- Reminders are organisational aids, not a clinical decision system. They must not suggest changing, stopping or doubling medicines.
- The project does not diagnose, prescribe, modify treatment or replace a clinician. Urgent symptoms require appropriate medical care.
- Caregiver sharing, external notifications, calendar access and storage must be opt-in and protected.

What we can and cannot claim

We can claim: CalmCare is designed to simplify discharge instructions, preserve source facts, make uncertainty visible, support selected languages and provide voice and print access. We can report measured results from our tests.

We should not claim without evidence: that no app anywhere has similar features, that the system prevents readmissions, that every translation is perfect, or that a high model score guarantees medical safety.

Final project definition

CalmCare is a multilingual, voice-first, safety-verified patient recovery companion that converts a discharge summary into a plain-language care plan, ties important instructions back to their source, and supports patients and caregivers in following the documented plan without inventing medical advice.

Prepared from the supplied H2 problem statement and our project discussions. Feature set and architecture are proposed design decisions; exact implementation depends on time, tools, language support, and testing.